

2-Phase Interleaved Sync-Buck Converter

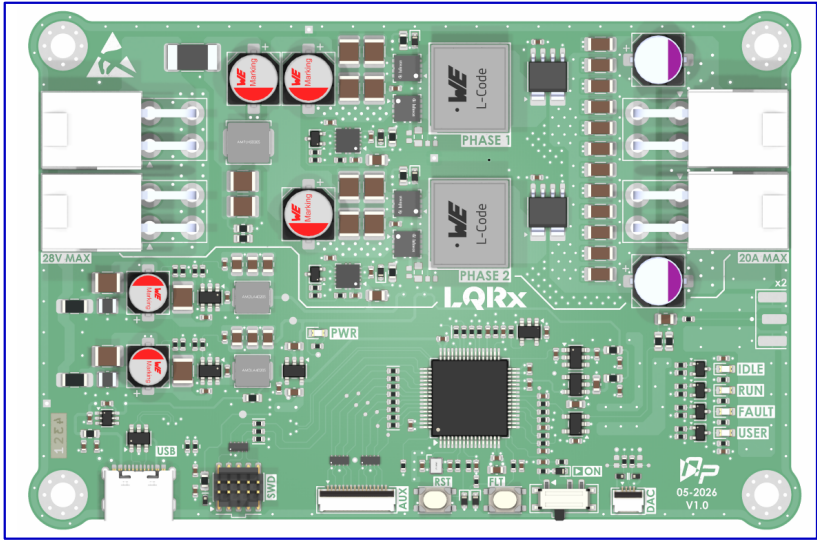
Digitally Controlled Power Conversion with STM32G474

Revision 1.1

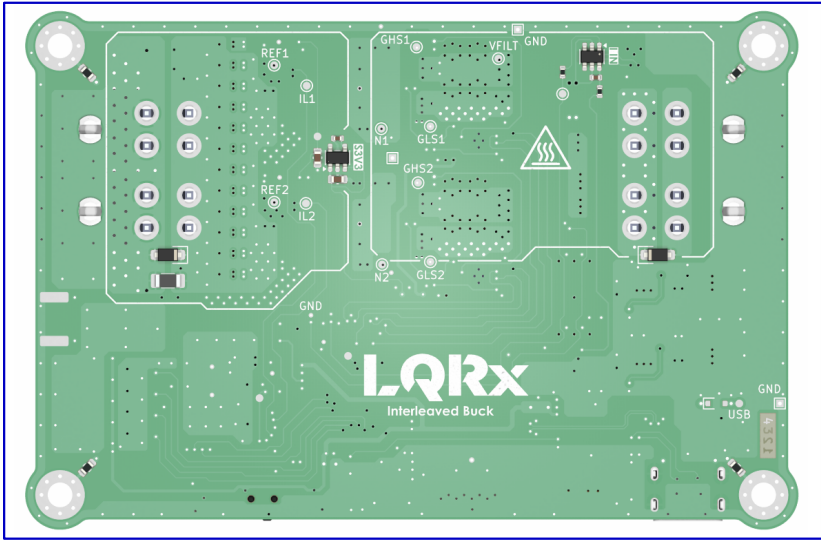
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TOP VIEW



BOTTOM VIEW



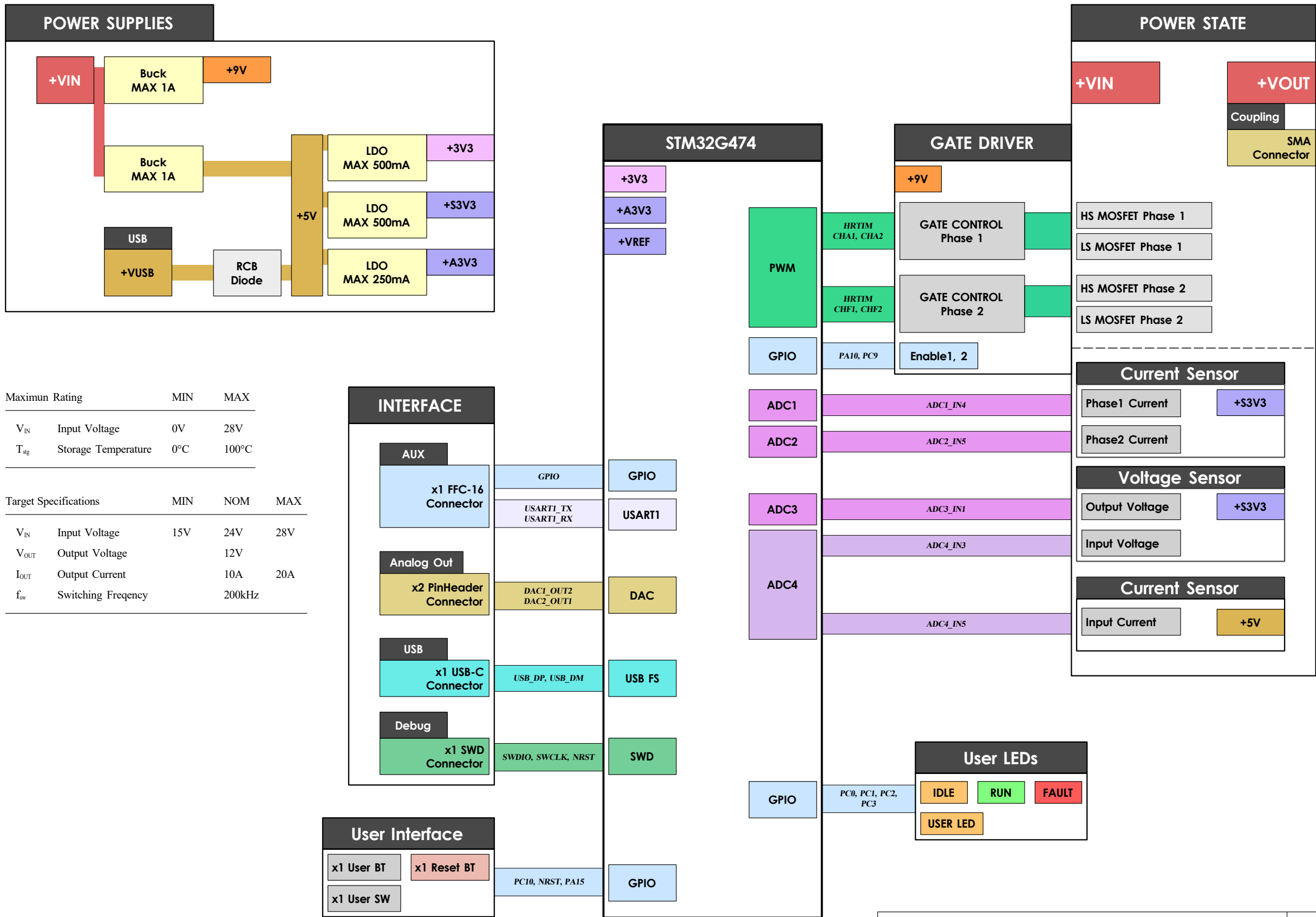
Legend

DESIGN NOTE: Text for information design.	DEBUG NOTE: Text for Debug notes.	DESIGN NOTE: Text for design guideline	DESIGN NOTE: Text for critical design	LAYOUT NOTE: Text for layout guideline	LAYOUT NOTE: Text for Critical layout	General Note Warning Note Text to be added to silkscreen.
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LQRx

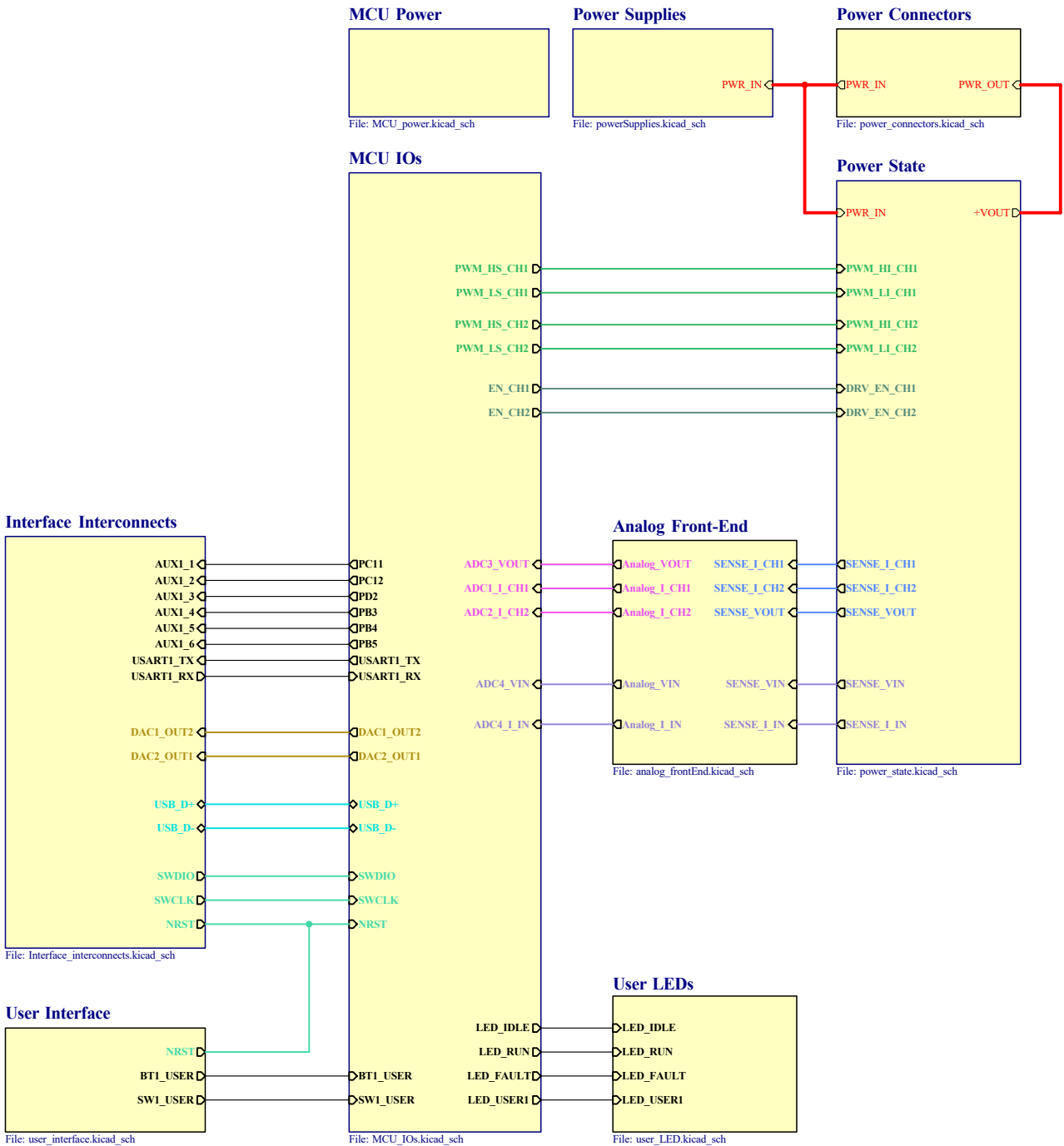
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Project Overview Cover Page	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /	Size: A3	Sheet: A of 9
KiCad E.D.A. 10.0.4		File: 2Phase_SynchBuck.kicad_sch	

[B] Block Diagram



Project: 2-Phase Interleaved Sync-Buck Converter			LQRx		
Title: Block Diagram (Top Level View)	Drawing by: NATCHPAI		Date: 08/04/2026	Variant:	
	Reviewed by:		Date:	Rev: 1.1	Date: 28/06/2026
	Sheet: /Block Diagram/			Size A3	Sheet:
	KiCad E.D.A. 10.0.4			File: Block_diagram.kicad_sch	
6		7		8	

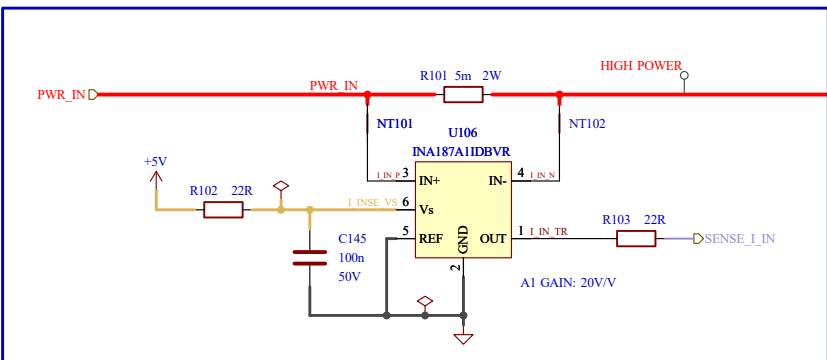
[C] Top Hierarchical View



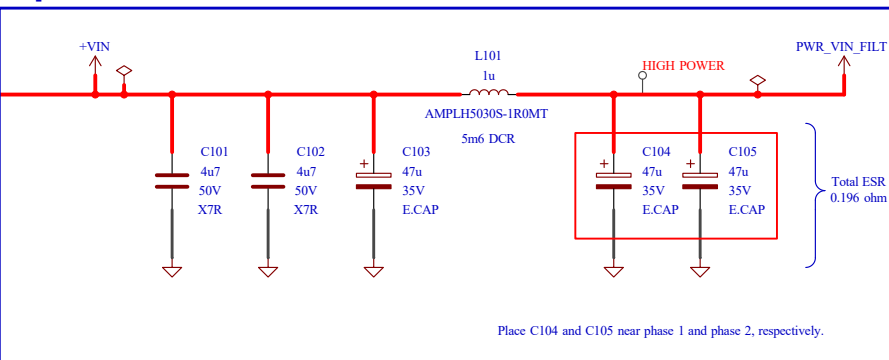
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Top Hierarchical View (Top Level View)	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/	Size: A3	Sheet: C of 9
KiCad E.D.A. 10.0.4		File: Top_hierarchical.kicad_sch	

[1] Power State

I_{IN} Sense

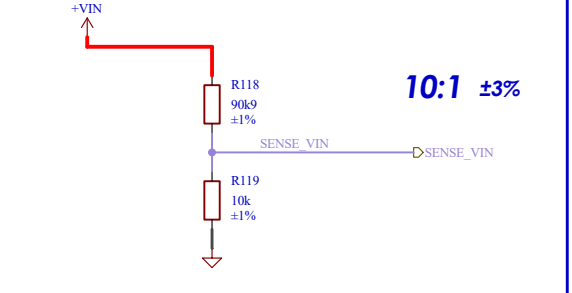


Input Filter

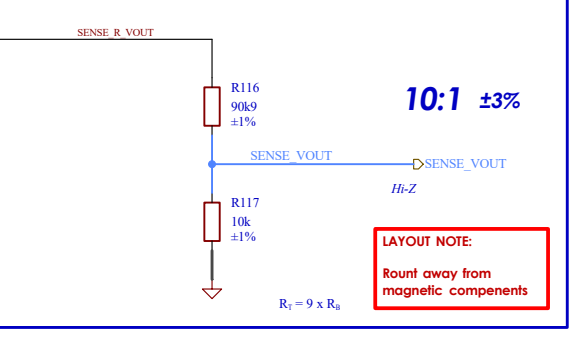


DESIGN NOTE:
GRT32ER71H475KE01
Input Capacitors 4.7uF x8 Derating -33% @24V = 25.2uF
 $C_d = 4^\circ C = 100\mu F$
 $Z_s = \sqrt{L/C} = \sqrt{1/25.2} = 0.199\Omega$
 $R_d = Z_s^2 / (L/Z_s - DCR)$
 $Q=0.9, DCR=15m \Rightarrow R_d = 0.192\Omega$
<Target 100uF ESR 0.192ohm>
Used Electrolytic Cap.(47uF, ESR 0.39ohm) x2 for stability
 $Z_{filter} Peak -13.78dB @28kHz$
<Atten. -20.5dB @ 200kHz

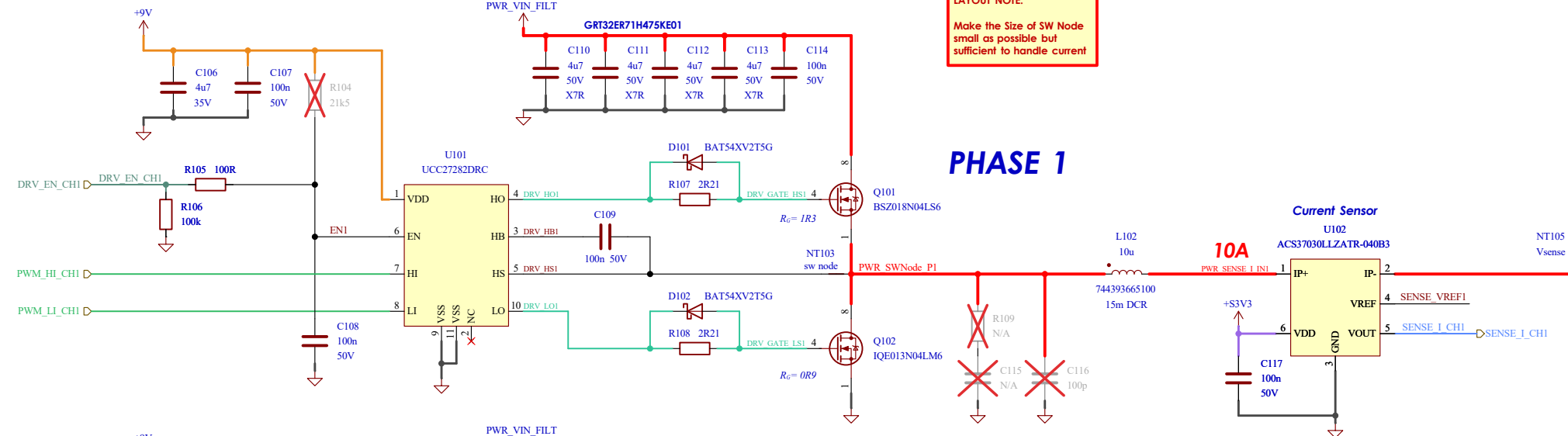
Input Voltage Sense



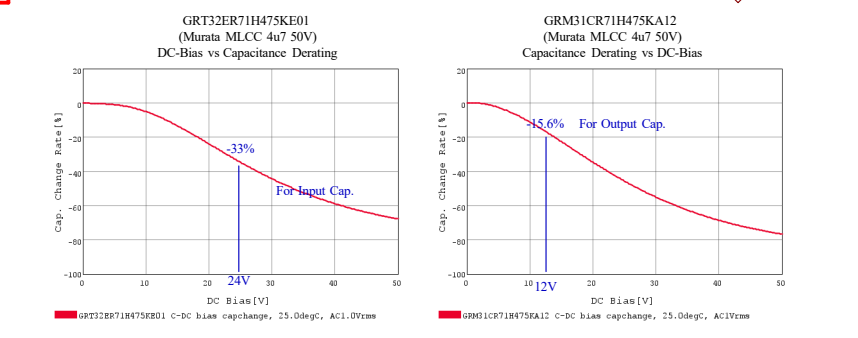
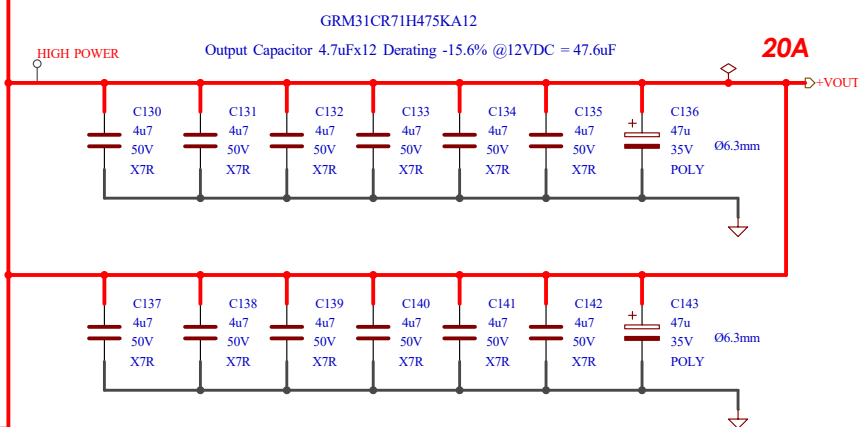
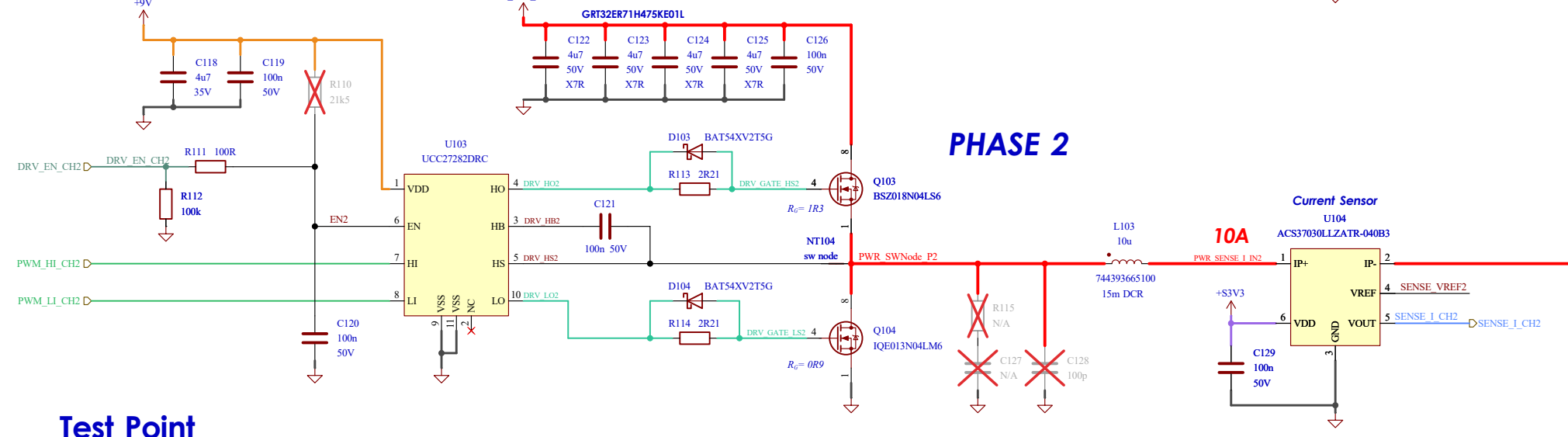
Output Voltage Sense



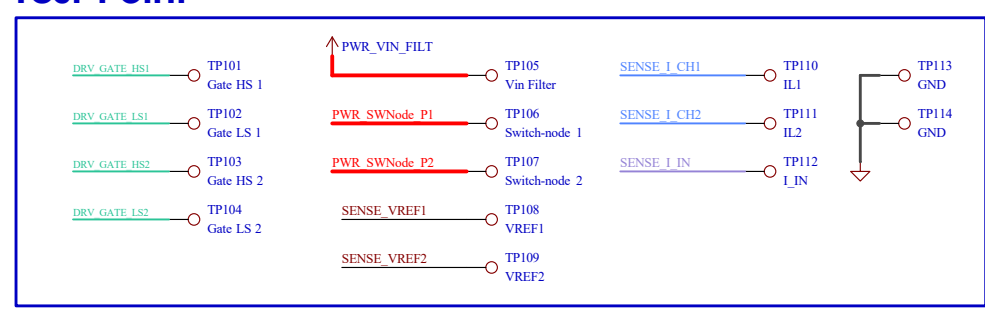
PHASE 1



PHASE 2



Test Point



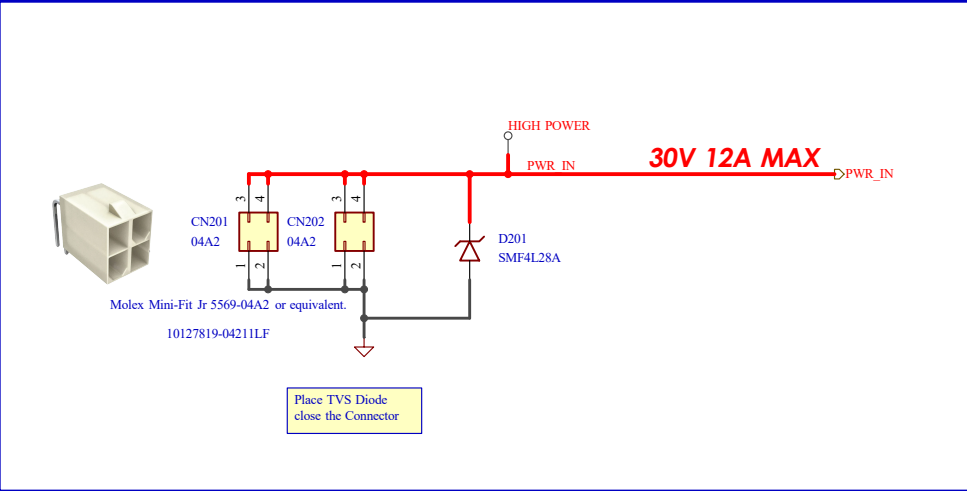
DEBUG NOTE:
If Switch-node are ringing
C116 and C128 are capacitor to find R C values in the snubber circuit.
 $C_{snr} = C_{gr} [f_2^2 / (f_1^2 - f_2^2)]$
 $R_{snr} = 1 / [2\pi f_1 C_{snr}]$
 $C_{snb} = (4 \cdot 8) C_{gr}$
 $R_{snb} = 1 / Q [\sqrt{L_{gr} / C_{snb}}]$



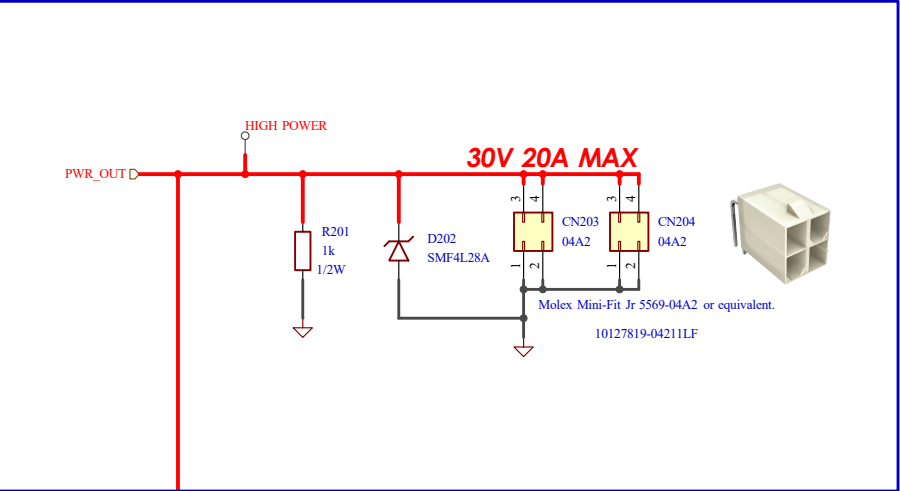
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power State 2-Phase Buck & Gate Driver	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
Sheet: /Top/Power State/ File: power_state.kicad_sch		Size: A3	Sheet: 1 of 9
KiCad E.D.A. 10.0.4			

[2] Power Connectors

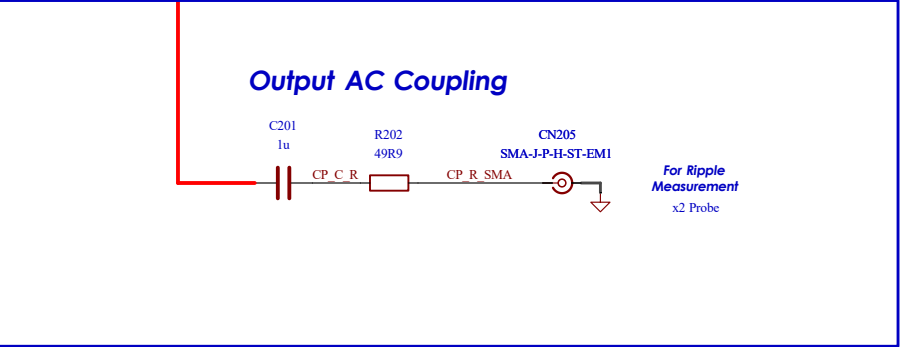
Input Connector



Output Connector



Output AC Coupling

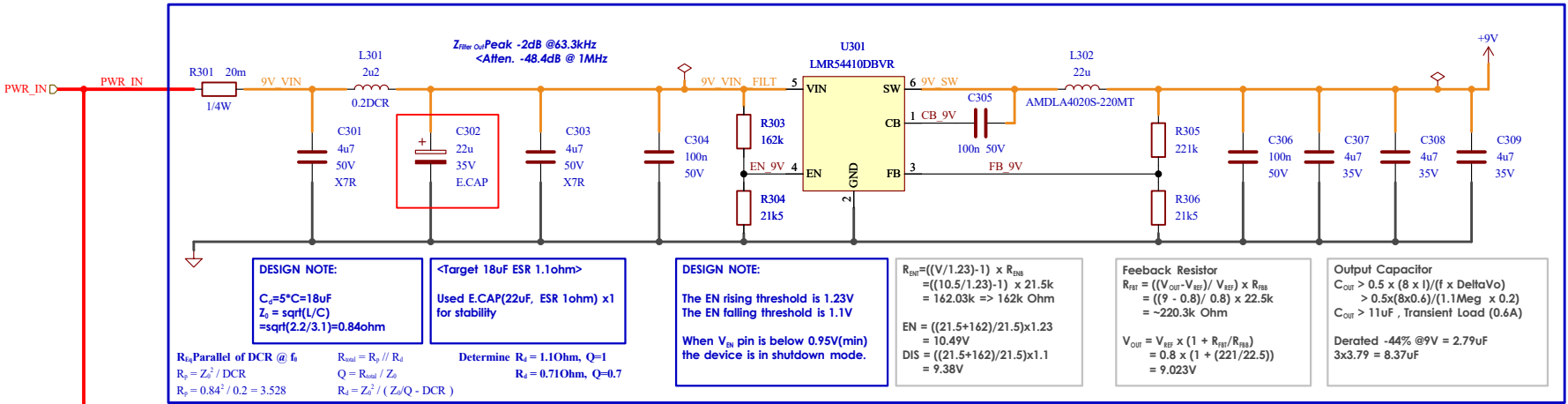


Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Connectors	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/Power Connectors/		Size A3
KiCad E.D.A. 10.0.4	File: power_connectors.kicad_sch		Sheet: 2 of 9

[3] Power Supplies

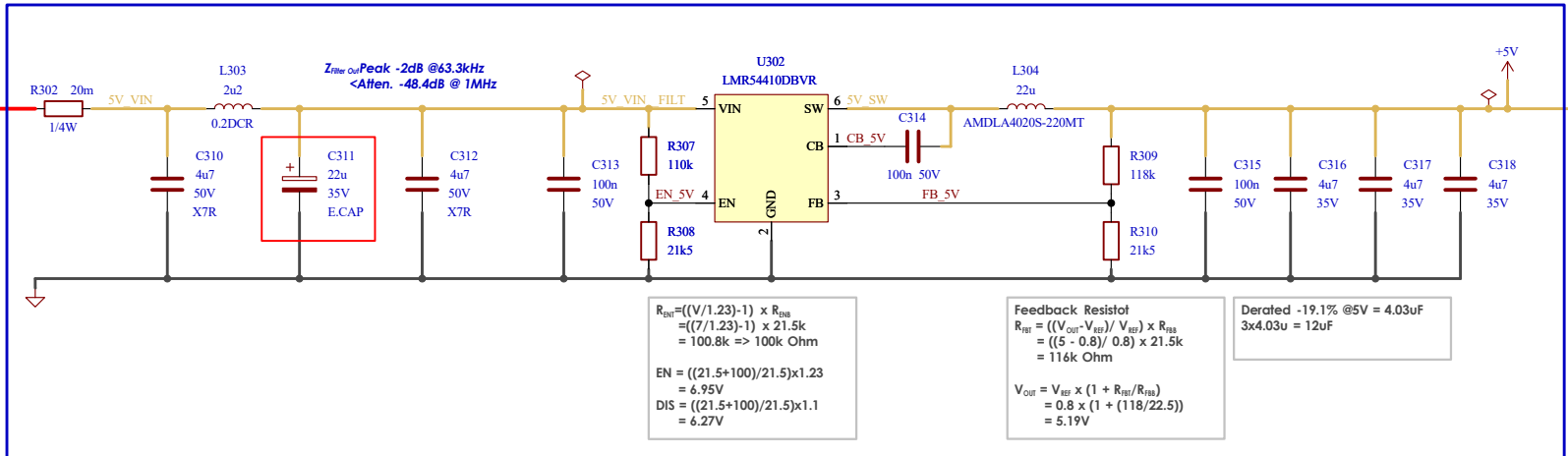
Buck Converter 9V 1A (1.1MHz)

*Enable when V_{IN} is above 11V

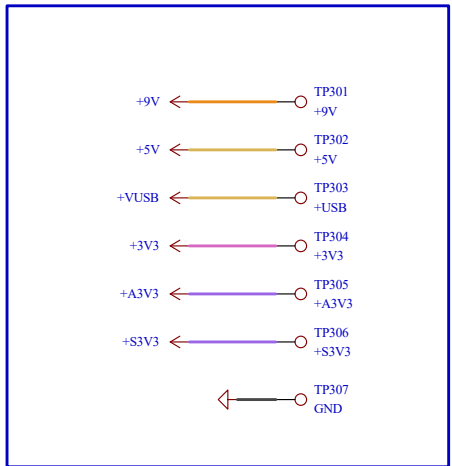


Buck Converter 5V 1A (1.1MHz)

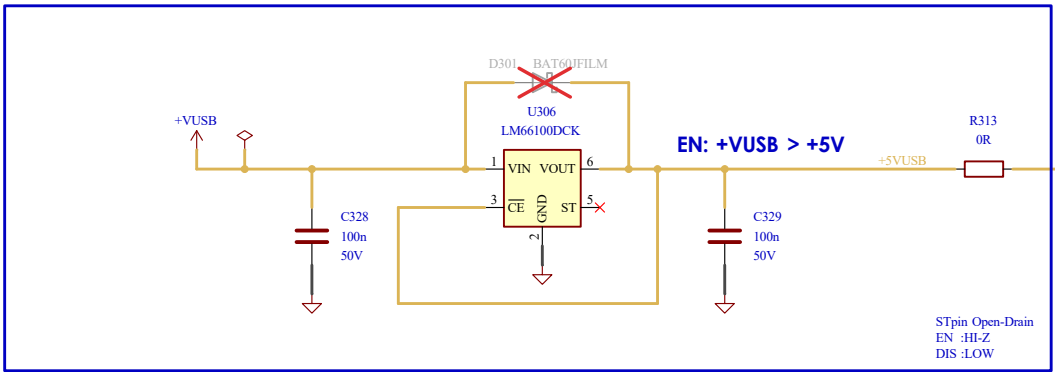
*Enable when V_{IN} is above 7V



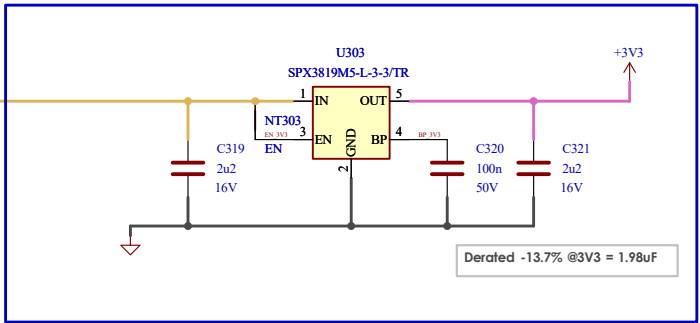
Test Point



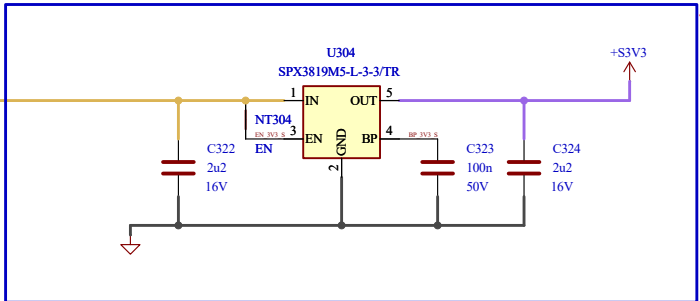
Reverse Current Blocking



LDO 3V3 500mA (Digital path)

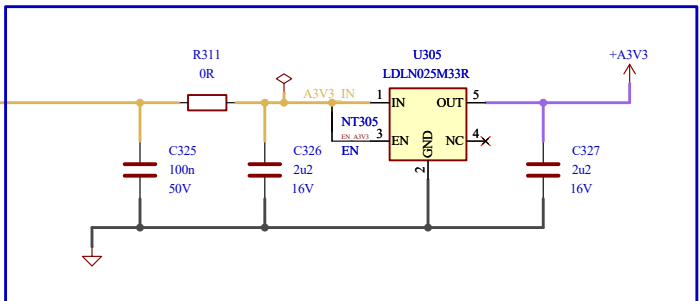


LDO 3V3 500mA (Sensor path)

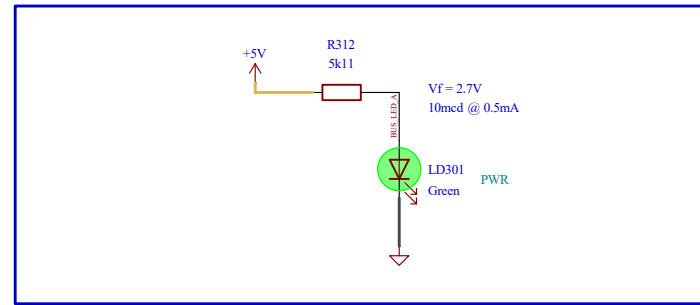


LAYOUT NOTE:
Place this power supply near the current sensor
On Page 1

LDO 3V3 250mA (Analog path)

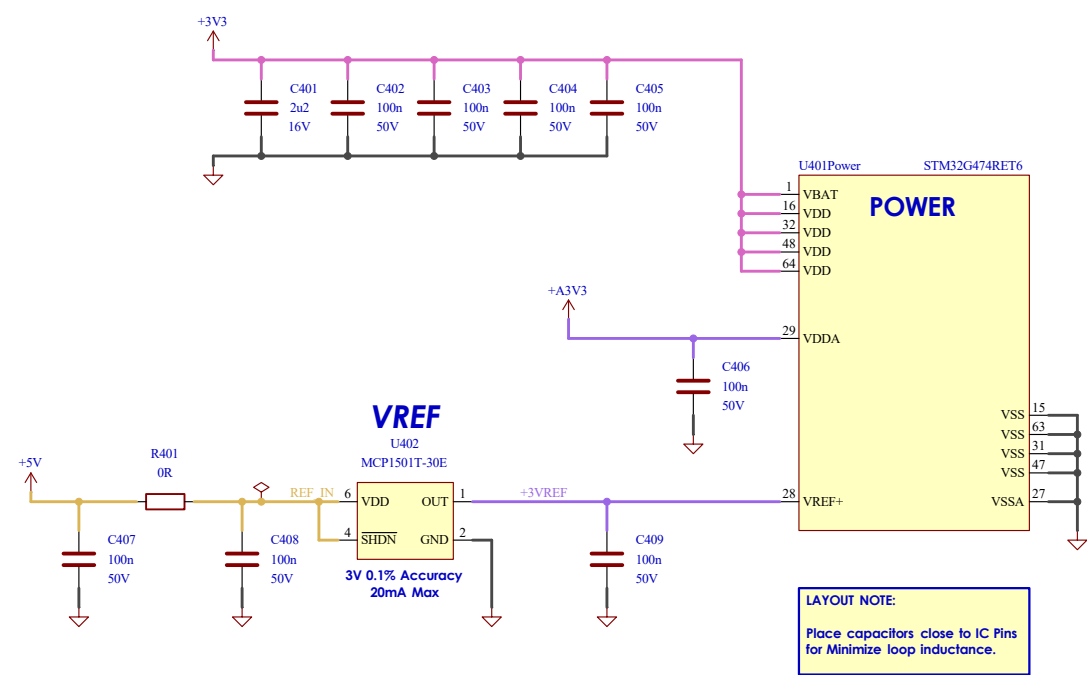


LED Indicators



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Supplies Auxiliary 9V 5V 3V3	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
KiCad E.D.A. 10.0.4	Sheet: /Top/Power Supplies/ File: powerSupplies.kicad_sch		Size: A3 Sheet: 3 of 9

[4] Microcontroller Power



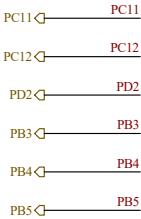
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Microcontroller Power STM32G474Rx	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/MCU Power/ File: MCU_power.kicad_sch		Size A3 Sheet: 4 of 9
KiCad E.D.A. 10.0.4			

[5] Microcontroller IOs

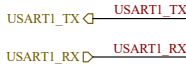
Interconnects

Auxiliary Port 1

GPIO



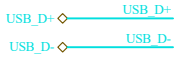
USART



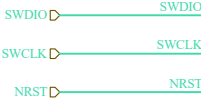
Analog Output (DAC)



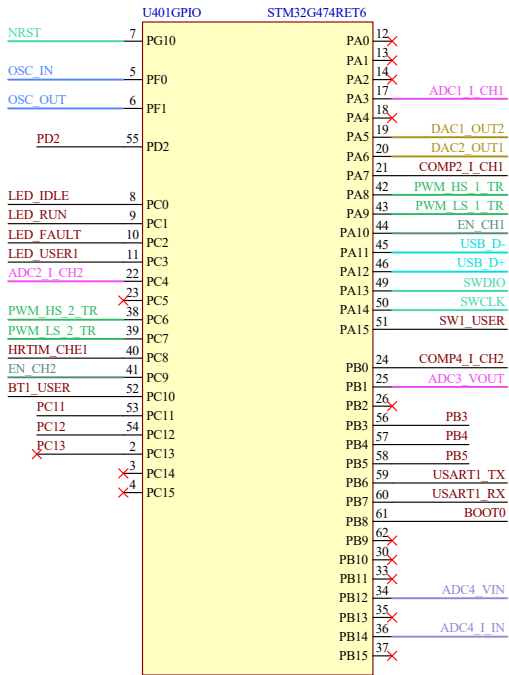
USB2.0 FS



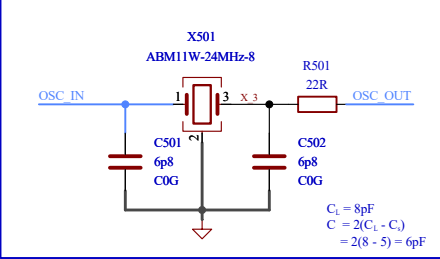
Serial Wire Debug



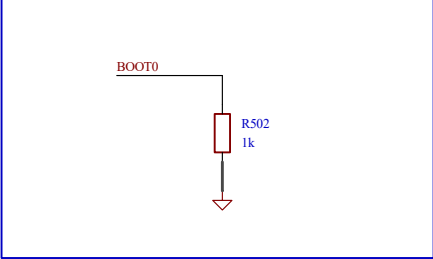
STM32G474 I/O



HSE Clock (24MHz)



BOOT0



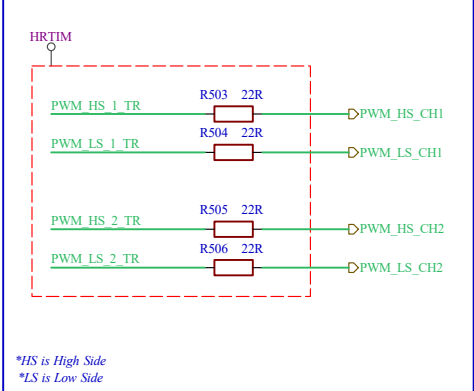
User Interface



Test Point



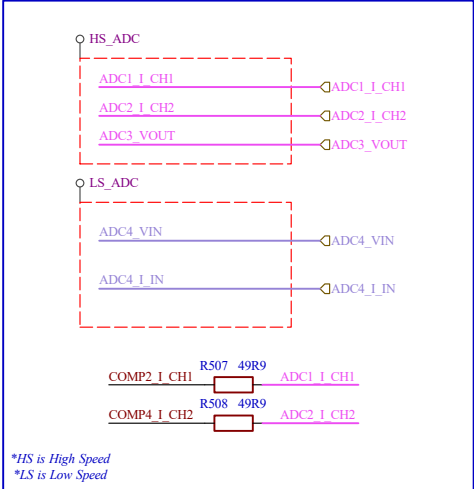
High Resolution Timer



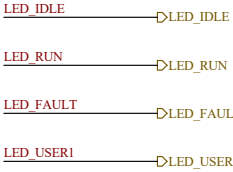
Enable Gate Driver



ADC



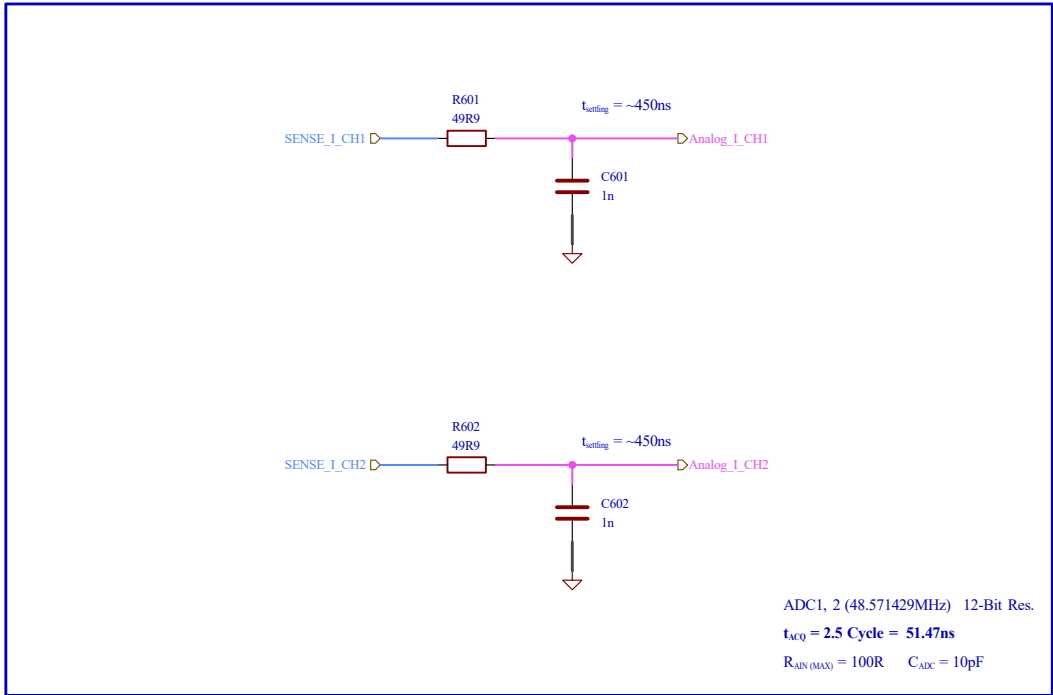
LEDs



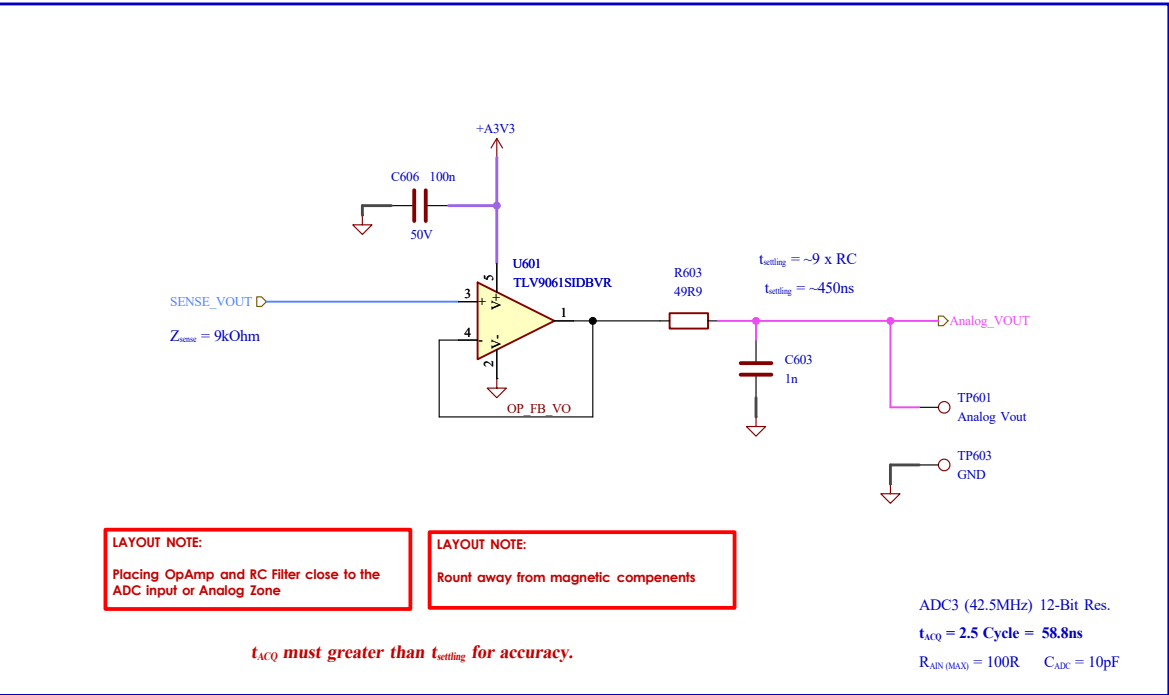
Project: 2-Phase Interleaved Sync-Buck Converter				LQRx			
Title: Microcontroller IOs STM32G474Rx		Drawing by: NATCHPAI		Date: 08/04/2026		Variant:	
		Reviewed by:		Date:		Rev: 1.1	Date: 28/06/2026
		Sheet: /Top/MCU IOs/ File: MCU_IOs.kicad_sch				Size A3	Sheet: 5 of 9
KiCad E.D.A. 10.0.4							

[6] Analog Front-End

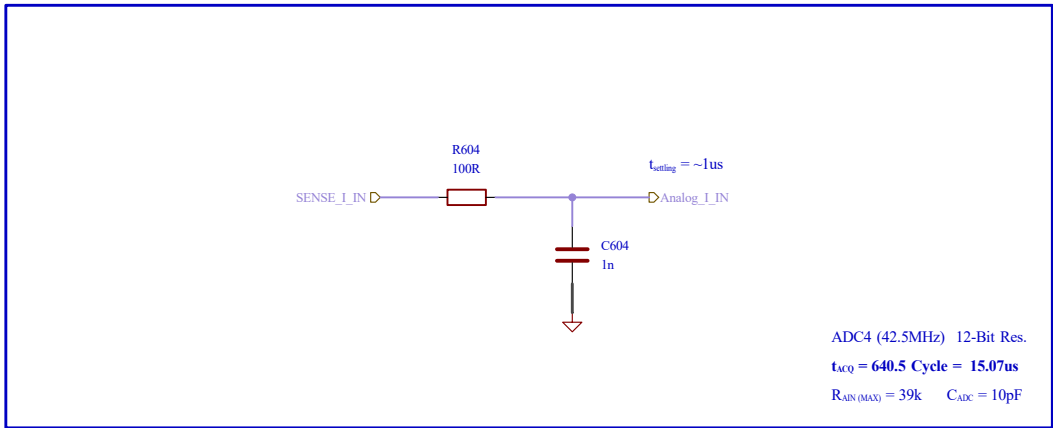
Current Path (High Speed)



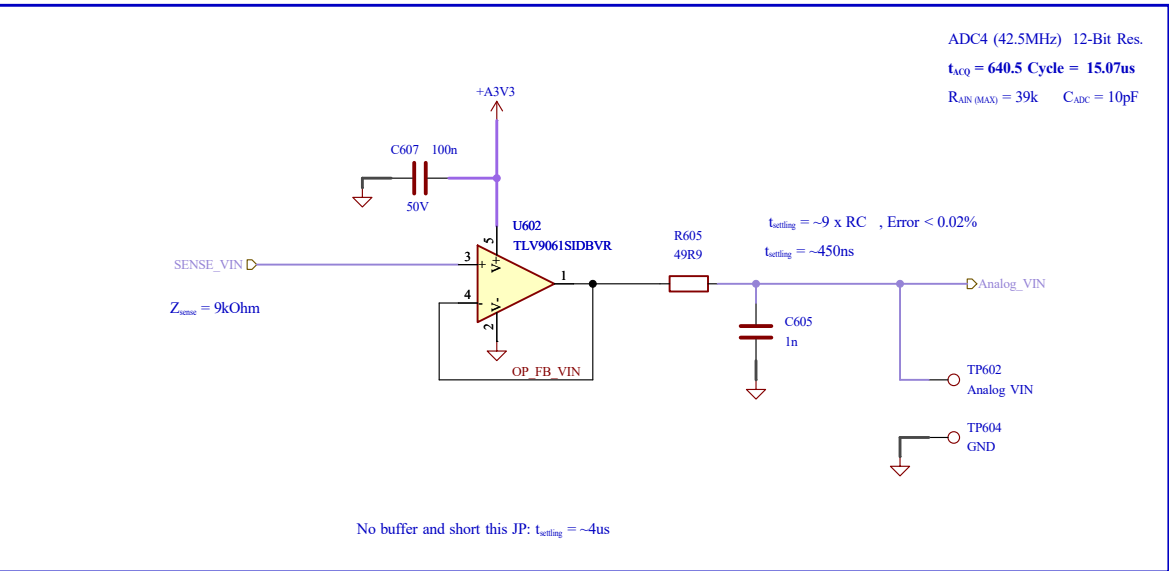
Output Voltage Path (High Speed)



Input Current Path (Low Speed)



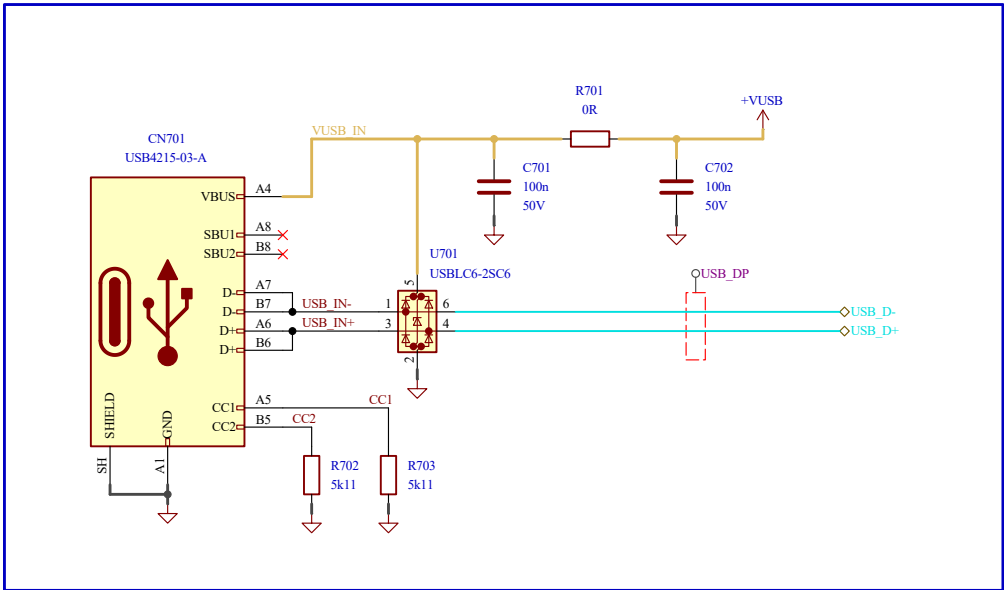
Input Voltage Path (Low Speed)



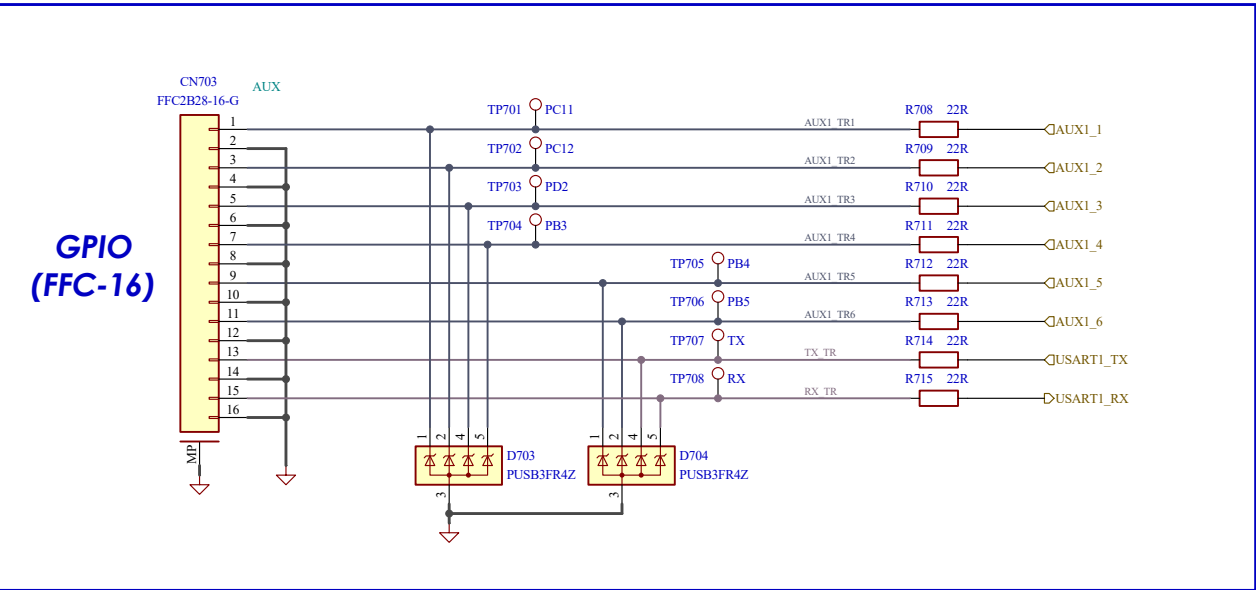
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Analog Front-End	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/Analog Front-End/ File: analog_frontEnd.kicad_sch		Size A3 Sheet: 6 of 9
KiCad E.D.A. 10.0.4			

[7] Interface Interconnects

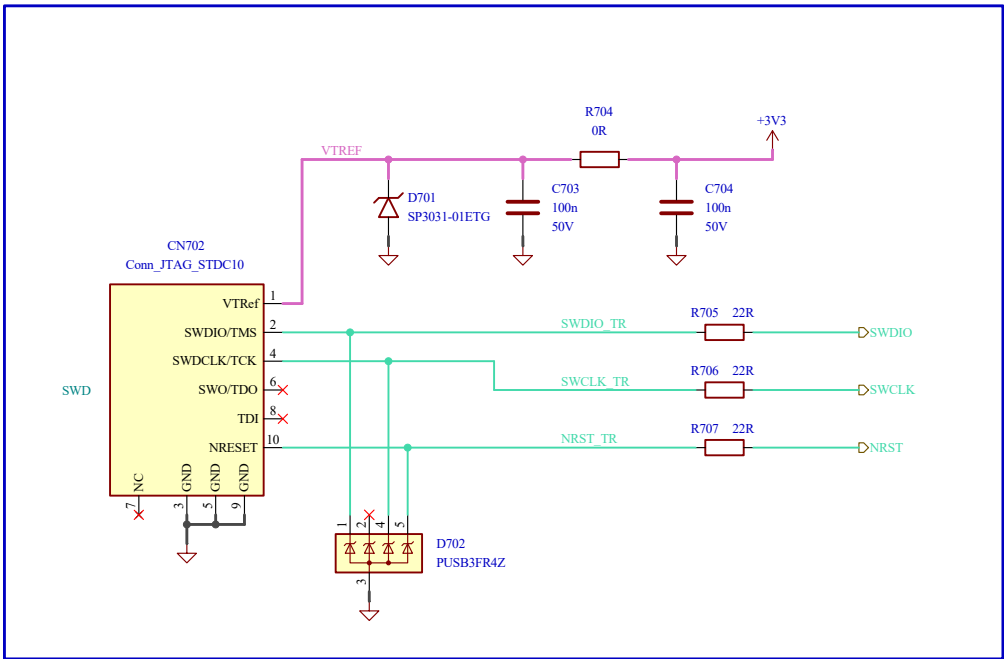
USB-C Connector



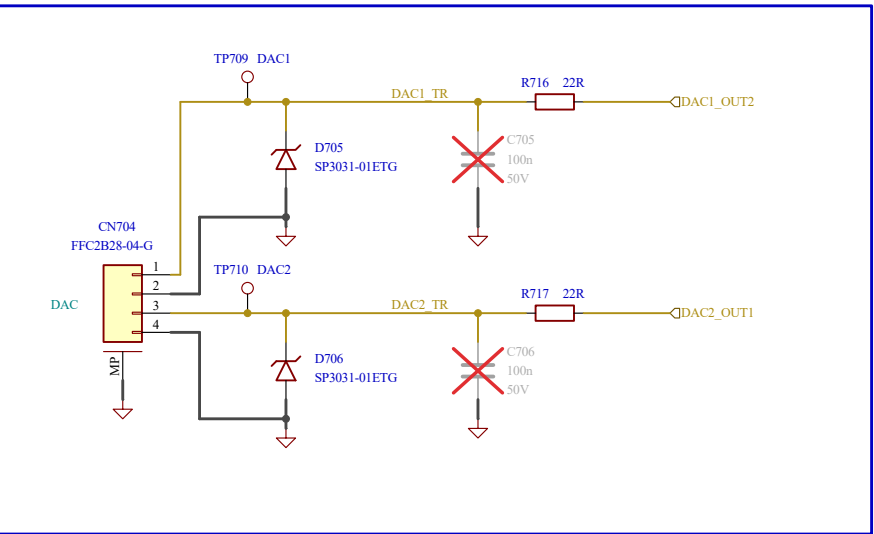
AUX Connector



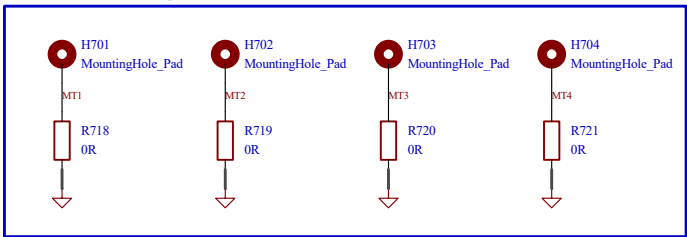
SWD Connector



Analog Output



Mounting Hole

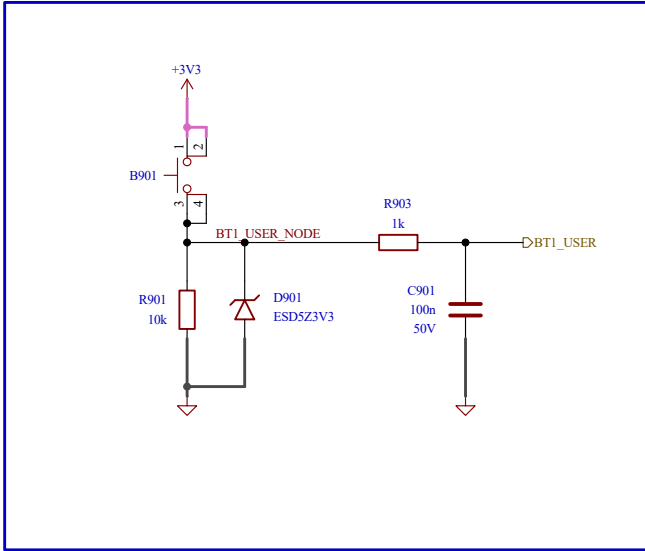


Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Interface Interconnects	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
Sheet: /Top/Interface Interconnects/		Size: A3	Sheet: 7 of 9
KiCad E.D.A. 10.0.4		File: Interface_interconnects.kicad_sch	

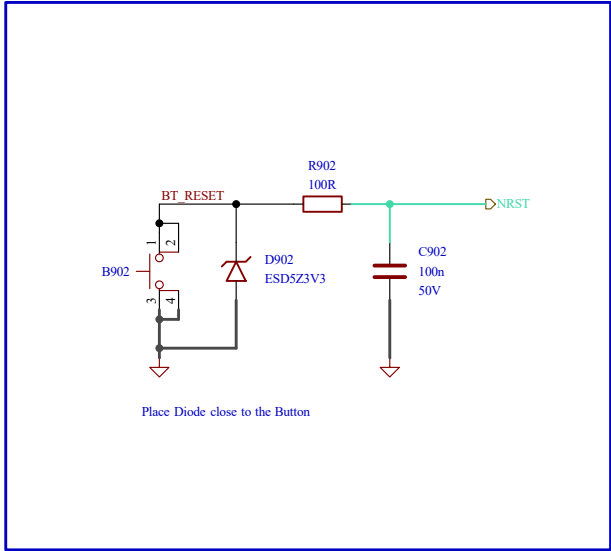
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: User LEDs Status LEDs	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/User LEDs/		Size A3 Sheet: 8 of 9
KiCad E.D.A. 10.0.4	File: user_LED.kicad_sch		

[9] User Interface

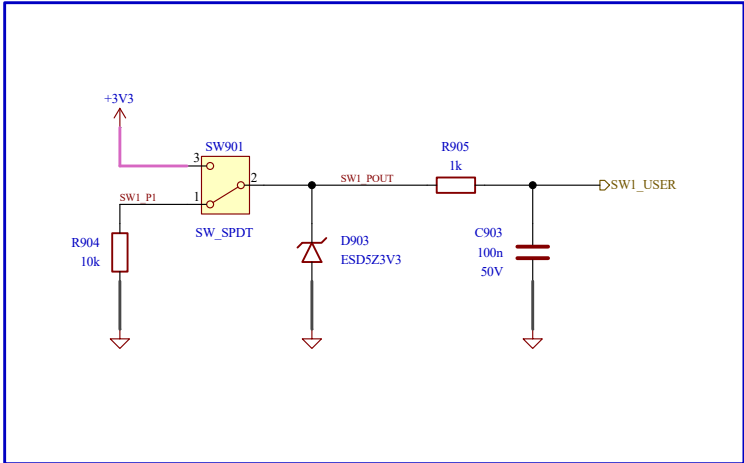
User Button



Reset Button

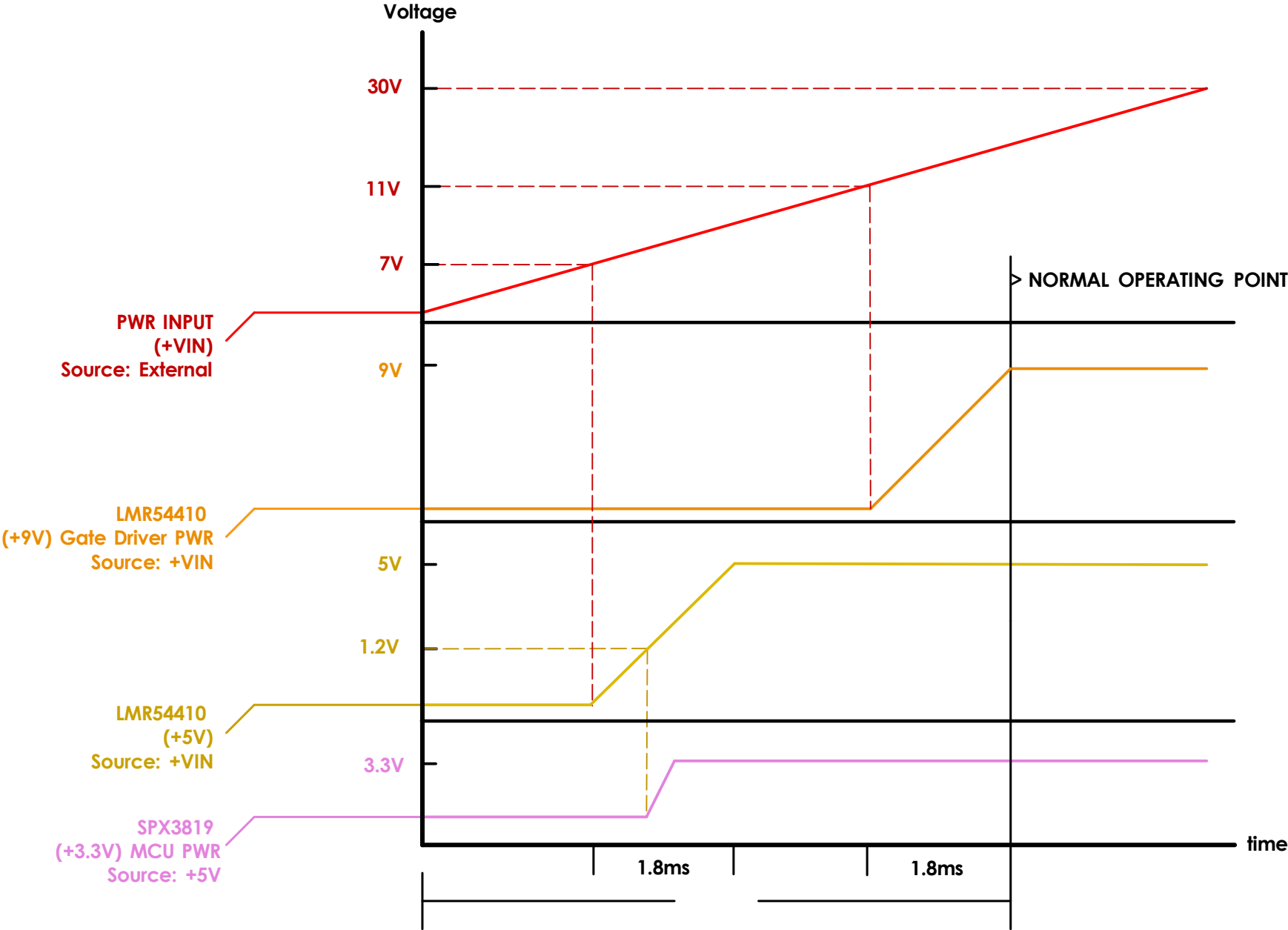


User Switch



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: User Interface Button & Switch	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/User Interface/ File: user_interface.kicad_sch		Size A3 Sheet: 9 of 9
KiCad E.D.A. 10.0.4			

[D] Power Supply Sequencing



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Power Supply Sequencing	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.1	Date: 28/06/2026
	Sheet: /Power Supply Sequencing/		Size: A3	Sheet: D of 9
KiCad E.D.A. 10.0.4		File: pwr_sequencing.kicad_sch		

[E] Revision History

Revision 1.0

- (18/05/2026) Initialize the project
- (03/06/2026) PC Board is Released

Revision 1.1

- (28/06/2026)
- Removed the Mosfet that Enable the ICs Gate Driver (p.1)
 - Changed ADC input Capacitor 47pF to 1nF. (p.6)

Project: 2-Phase Interleaved Sync-Buck Converter				LQRx	
Title: Revision History	Drawing by: NATCHPAI		Date: 08/04/2026	Variant:	
	Reviewed by:		Date:	Rev: 1.1	Date: 28/06/2026
	Sheet: /Revision History/			Size A3	Sheet: E of 9
KiCad E.D.A. 10.0.4		File: RevHistory.kicad_sch			